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Virulence gene mutations as a differentiator of clinical phenotypes: insights from community-acquired uropathogenic *Escherichia coli*

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Abstract

Uropathogenic *Escherichia coli* (UPEC) remains an important cause of urinary tract infection during pregnancy. Multiple molecular virulence determinants and antibiotic resistant genes facilitate its pathogenesis and virulence phenotype. Hence it is hypothesized that there will be considerable variation in genes among the isolates from symptomatic as well as asymptomatic bacteriuria (ABU) during pregnancy. The aim of this study was to decipher the genetic variation among the two phenotypes. Six different UPEC isolates collected from urine specimens of consecutive pregnant females (five, symptomatic bacteriuria and one, ABU) were tested for their growth kinetics, and biofilm formation. A total of 87 virulence determinants and 56 antibiotic resistance genes were investigated using whole-genome sequencing, to identify putative drivers of virulence phenotype. In this analysis, we identified eight different types of fully functional toxin antitoxin (TA) systems [HipAB, YefM-YoeB, YeeU-YeeV (CbtA), YhaV-PrfF, ChpBS, HigAB, YgiUT and HicAB] in the isolates from symptomatic bacteriuria; whereas partially functional TA system with mutations were observed in the asymptomatic one. Isolates of both the groups showed equivalent growth characteristics and biofilm-formation ability. Genes for an iron transport system (Efe UOB system, Fhu system except FhuA) were observed functional among all symptomatic and asymptomatic isolates, however functional mutations were observed in the latter group. Gene YidE was observed predominantly associated with the biofilm formation along with few other genes (BssR, BssS, YjgK, etc.). This study outlines putative critical relevance of specific variations in the genes for the TA system, biofilm formation, cell adhesion and colonization among UPEC isolates from symptomatic and asymptomatic bacteriuria among pregnant women. Further functional genomic study in the same cohort is warranted to establish the pathogenic role of these genes.

INTRODUCTION

Urinary tract infection (UTI) remains one of the most common complications during pregnancy. Women attending antenatal clinic may not necessarily present with any urinary symptoms, but mostly encounter various conditions effecting urinary tract such as asymptomatic bacteriuria (ABU), cystitis, vaginitis, pyelonephritis etc. or its complications [1]. The structural, functional and hormonal changes that happen in the body during gestation make pregnant women more susceptible to the above infections [2]. Poor immune response during pregnancy further increases the risk of infection. Bacteriuria among pregnant females without proper treatment may

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Keywords: urinary tract infection (UTI); pregnancy; UPEC (uropathogenic *Escherichia coli*); symptomatic bacteriuria; asymptomatic bacteriuria.

Abbreviations: ABU, Asymptomatic Bacteriuria; AMR, Antimicrobial Resistance; CFU, Colony Forming Unit; CHC, Community Health Center; CLED, cysteine lactose electrolyte deficient; MCP-3, methyl-accepting chemotaxis protein III; MHB, Mueller Hinton Broth; PHC, primary health center; RHC, Rural Health Center; SEM, Standard Error Mean; TAS, Toxin-Antitoxin System; TFP- A, Type-1 fimbrial protein A chain; UPEC, uropathogenic *Escherichia coli*; UTI, urinary tract infection.

†These authors contributed equally to this work

Two supplementary tables are available with the online version of this article.

lead to development of more resistant bacteria and pose serious threats to mother and foetus causing complications like pyelonephritis, amnionitis, hypertension, low birth weight, still birth, preterm labour, pre-eclampsia, etc. [3, 4]. Moreover, controversies regarding screening and treatment of asymptomatic bacteriuria in pregnancy are documented, which validate the need of more in-depth research on the same [5].

Escherichia coli remains the most common uropathogen accounting approximately 80% of total UTI and found in both symptomatic as well as asymptomatic conditions [6]. It is hypothesized that ABU is probably caused by the commensal *E. coli* followed by the evolution of pathogenic *E. coli* by various gene acquisition, transformation of virulence factors and biofilm formation [7]. Various molecular changes may lead to difference in clinical features, which can only be explained by an in-depth molecular analysis of the causal bacterial isolate. Hence, the objective of this paper was to compare and determine the variety of virulence determinants and antibiotic resistance genes among the *E. coli* isolates from pregnant females attending rural primary healthcare facilities of India presenting with either symptomatic or asymptomatic bacteriuria.

METHODS

Participants

This study is a part of ongoing multi-centric on community-acquired UTI and emerging drug resistance (CAUTION-ED) conducted prospectively from October 2019 to September 2020 at different primary healthcare centres of India. Ethical approval was obtained from the institute ethical committee, and informed consent was obtained from every patient included in the study. As per the protocol of the study, clean-catch mid-stream urine samples were collected from the pregnant females attending the antenatal clinic of primary health centre (PHC) and Community Health Center (CHC) located in rural area.

Inclusion criteria

Pregnant women with clinical diagnosis of cystitis, uncomplicated UTI (without any anatomical or functional abnormality), recurrent UTI or asymptomatic bacteriuria were included in the study.

Exclusion criteria

Pregnant women with a history of hospitalization 1 week prior to the presentation, neurogenic bladder, obstructive uropathy and if on antibiotic prophylaxis, were excluded from the study.

Clinical samples

Clean catch midstream urine sample was obtained from the pregnant females in the antenatal clinic. The samples were processed as per the standard operating procedure. Urine specimen was cultured on cysteine lactose electrolyte deficient (CLED) media and incubated at 37°C overnight. Growth equivalent to more than equal to 10^5 c.f.u. ml⁻¹ was considered significant.

Bacterial isolates and cultures

The six study isolates from different pregnant females were identified as *E. coli* using MALDI-TOF MS (VITEK-MS system, BioMérieux, Marcy-l'Étoile, France). The antimicrobial susceptibility testing was done using VITEK II (BioMérieux, Marcy-l'Étoile, France). Since all the participants belong to Out Patient Department (OPD), the oral antibiotics of the UTI panel were used for AST testing. The tested antibiotics were, amikacin, amoxy-clavulanic acid, cefalotin, cefoxitin, cefixime, ceftazidime, ceftriaxone, gentamycin, ciprofloxacin, ofloxacin, trimethoprim/sulphamethoxazole, nitrofurantoin, fosfomycin and ertapenem. MIC was determined and susceptibility was interpreted using the CLSI 2020 guideline. All microbial isolates were stored at -80°C in glycerol stock solution. Further, the isolates were analysed for growth kinetics, biofilm formation, virulence marker genes and antimicrobial resistance gene by whole genome sequencing.

Growth kinetics

Growth kinetics to establish the relationship between growth rate and nutrients present in the isolates were done as per universally practiced methods [8]. In brief, isolates retrieved from glycerol stock stored at -80°C in Mueller-Hinton broth (MHB) and incubated at 37°C with continuous shaking at 250 r.p.m. overnight. Then, 10^4 cells ml⁻¹ were incubated as a secondary culture at 37°C with shaking at 250 r.p.m. Cells ml⁻¹ was counted by using OD₆₀₀ at Ultrospec 10 cell density metre (Biosciences, Buckinghamshire, UK) at different time points till 24 h and a growth curve was plotted with SEM of three independent biological replicates, each with three technical replicates.

Biofilm formation

Biofilm formation ability was checked by staining with crystal violet (C0775, Sigma, MO, USA) of three replicates each in three independent experiments. The culture from glycerol stock was retrieved in MHB and incubated overnight at 37°C with shaking at

Table 1. Demographic and clinical profile of the patients

Profile of patient	P10188(S)	P10192(S)	P10243(A)	P10300(S)	P10301(S)	P10405(S)
Age	19	21	28	20	19	30
Occupation	Housewife	Housewife	Housewife	Housewife	Housewife	Housewife
Education	Secondary	Secondary	Primary	Primary	Primary	Primary
Pregnancy (trimester)	second	first	first	first	first	third
Socioeconomic status	Poor	Poor	Poor	Poor	Poor	Poor
Fever without focus	Present	Present	Absent	Absent	Present	Absent
Turbid urine	Absent	Present	Absent	Absent	Present	Absent
Malodorous urine	Present	Present	Absent	Absent	Present	Absent
Hematuria	Absent	Absent	Absent	Present	Absent	Absent
Dysuria	Present	Present	Absent	Absent	Present	Absent
Frequency	Present	Present	Absent	Present	Present	Present
Urgency	Present	Present	Absent	Present	Absent	Present
Suprapubic pain	Present	Present	Absent	Present	Absent	Present

250 r.p.m. The subculture was grown till mid-log phase, and 10^6 cells ml^{-1} were inoculated in the 24-well plate (142475, Thermo, MA, USA). After 48 h incubation at 37°C , supernatant was discarded and washed twice with sterile PBS buffer to remove loosely attached cells. The attached cells were stained with 0.5% crystal violet followed by washing twice with PBS. Then it was dissolved by using 95% ethanol and reading was taken at absorbance of the OD595 ELISA reader (Biorad Hercules, CA, United States).

DNA extraction for molecular analysis

In total, 5 μl of glycerol stock of each bacterial strain was revived in 5 ml (1:1000) of MHB media (HiMedia, Mumbai, India), for overnight at 37°C temperature and 250 r.p.m. and sub-cultured in 5 ml MHB media in 1:100 for 3 h at 37°C and 250 r.p.m. DNA was isolated by a commercially available DNA isolation kit (Real Biotech Corporation, Taipei, Taiwan) according to the manufacturer's instructions. Briefly, 10^8 cells were mixed in 200 μl lysis buffer-1 and incubated it for 5 min at room temperature. Next, 200 μl cell lysis buffer-2 was added and incubated at 70°C for 10 min. Finally, 200 μl ethanol was added to prepare the lysate for binding and transfer it through the column. DNA was eluted using 70 μl pre-warm DNA elution buffer.

DNA quality control, library preparation and sequencing

Genomic DNA (gDNA) was quantified using a DS-11 Spectrophotometer (Denovix, Wilmington, USA), Qubit hsDNA assay (Thermo Scientific, USA) and checked for its integrity using agarose gel electrophoresis. Library preparation was performed at Genotypic Technology's Genomics facility using the plexWell WGS-24 Library Preparation protocol. Briefly, 5 ng genomic DNA was fragmented (fragmented and adaptor tagged) and uniquely barcoded. This was amplified by PCR, purified using MAGwise Beads (MG5000) and quantified by using a Qubit fluorometer. Then, libraries were prepared using paired-end sequenced on an Illumina HiSeq X Ten sequencer (Illumina, San Diego, USA) for 150 cycles following the manufacturer's instructions.

Data analysis

Libraries were sequenced using Illumina Hiseq (150 $\times$ 2 chemistry). The low-quality reads were trimmed using Trim Galore [9]. The adapter clipped, high-quality reads were aligned against reference *E. coli* 536, *E. coli* CFT073, *E. coli* UTI89, *E. coli* K12, using the Bowtie2 [10] alignment tool. Variants were identified using samtools [11] and bcftools [12]. The generated variants were filtered using vcftools annotation was carried out for all the identified variants using Snpeff [13]. Complete-genome sequence of all study isolates was submitted to NCBI and the BioProject ID is PRJNA749142.

Genetic loci for virulence and antibiotics' resistance-gene analysis

Genes related to virulence and antimicrobial resistance were checked for the insertion and deletion mutation with the reference of *E. coli* 536, *E. coli* CFT073, *E. coli* K12 and *E. coli* UTI89. Overall, 56 genes related to antibiotic resistance and 87 related to virulence genes were checked in all isolates.

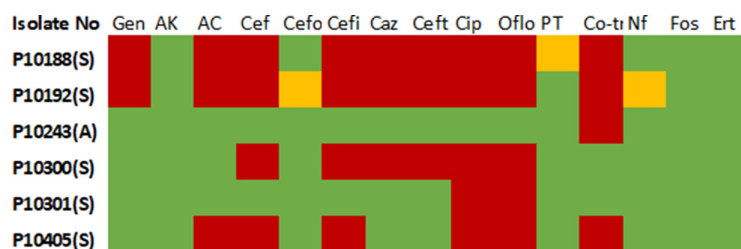


Fig. 1. Heat map of antimicrobial susceptibility test profile of symptomatic and asymptomatic isolates. Susceptible, intermediate and resistance. **Gen: gentamicin, Ak: amikacin, AC: amox/clavulanic acid, Cef: cefalotin, Cefo: cefoxitin, Cefi: cefixime, Caz: ceftazidime, Ceft: ceftriaxone, Cip: ciprofloxacin, Oflo: ofloxacin, PT: piperacillin/tazobactam, Co-tri: cotrimoxazole, Nf: nitrofurantoin, Fos: fosfomycin, Ert: ertapenem, (S) symptomatic, (A) asymptomatic.

Statistical analysis

The significance of data was analysed by using two-way ANOVA followed by Dunnett's multiple comparisons test by using GraphPad Prism version 6.0.0 for Windows, (GraphPad Software, San Diego, CA, USA, www.graphpad.com).

RESULTS

Participants and bacterial isolates

Six UPEC isolates from pregnant females were included in this study of which, five were from symptomatic (5/6, isolate no: P10188, P10192, P10300, P10301 and P10405) and one from asymptomatic bacteriuria (1/6, isolate no: P10243). Clinical profile of the participants was recorded during the enrolment process. Antibiotic susceptibility testing and whole-genome-sequence analysis data were compared between both the groups. Symptoms such as urgency, polyuria, are commonly manifested during pregnancy due to physiological changes in the body, thereby making it difficult for the clinician to suspect it as UTI. Therefore, in the current study fever, burning micturition, turbid and/or malodorous urination were considered as signs and symptoms for the detection of symptomatic UTI. *E. coli* remains the most common uropathogen in both symptomatic and asymptomatic bacteriuria conditions. Cases of symptomatic UTI are treated using empirical antimicrobials; whereas it remains unwarranted in asymptomatic UTIs except conditions like pregnancy or before any invasive urologic procedure. Moreover, the asymptomatic bacteriuria has been treated depending on the phylogenetic group of the microbes, virulence genes involved, pathogenic island, properties of biofilm formation and iron transport mechanism as documented earlier [14–16]. As previous studies reported, evolution of pathogenic *E. coli* happened from commensal *E. coli* due to horizontal transfer of many genes [17, 18]. It has been proven earlier that P fimbrial genes and iron uptake system encoding genes have a major role in effective colonization and survival of *E. coli* isolates in the urinary tract [19].

Clinical profile of the patients

Six out of thirty isolates were collected from pregnant women over a period of 4 months (5: causing symptomatic bacteriuria; 1: causing asymptomatic bacteriuria) (Table 1). All the participants were from CHC of India. The age distribution of the patients was between 19–30 years. (average: 22.8 years). Four participants were in their first trimester and one each were in the second trimester and third trimester. All the participants belong to lower socioeconomic status (as per modified Kuppuswamy scale). Comparing the signs and symptoms; fever, dysuria, malodorous and turbid urine were present 50, 50, 40 and 40% of the total patient, respectively. There was no history of diabetes, recurrent UTI, any urinary tract abnormality among any of the participants. Two out of six had a history of multiple pregnancies. In the current study, apart from the presentation of symptoms, there were no significant differences in any of the clinical or microbiological analysis. Fever was the most important symptom and more than 60% of the patients were in the first trimester of pregnancy.

Microbiological characteristics of the isolates

All six isolates had a significant colony count of $\geq 10^5$ CFU ml⁻¹ in the CLED agar and were identified as *E. coli* using MALDI-TOF. The antimicrobial susceptibility test result was given in Fig. 1. The isolates were tested for AST by using the VITEK 2 system. Among the six isolates 66.6% (4/6) were extended spectrum beta lactamase (ESBL) producers. The resistance to fluoroquinolone, co-trimoxazole, aminoglycoside, beta-lactam (BL) and beta-lactamase inhibitor (BL/BLI), third-generation cephalosporins, and carbapenem were 83.3% (5/6), 66.6% (4/6) and 33.3% (2/6), 100% (6/6), 50% (3/6), 66.6% (4/6) and 16.6% (1/6), respectively. All the isolates were susceptible to fosfomycin and nitrofurantoin.

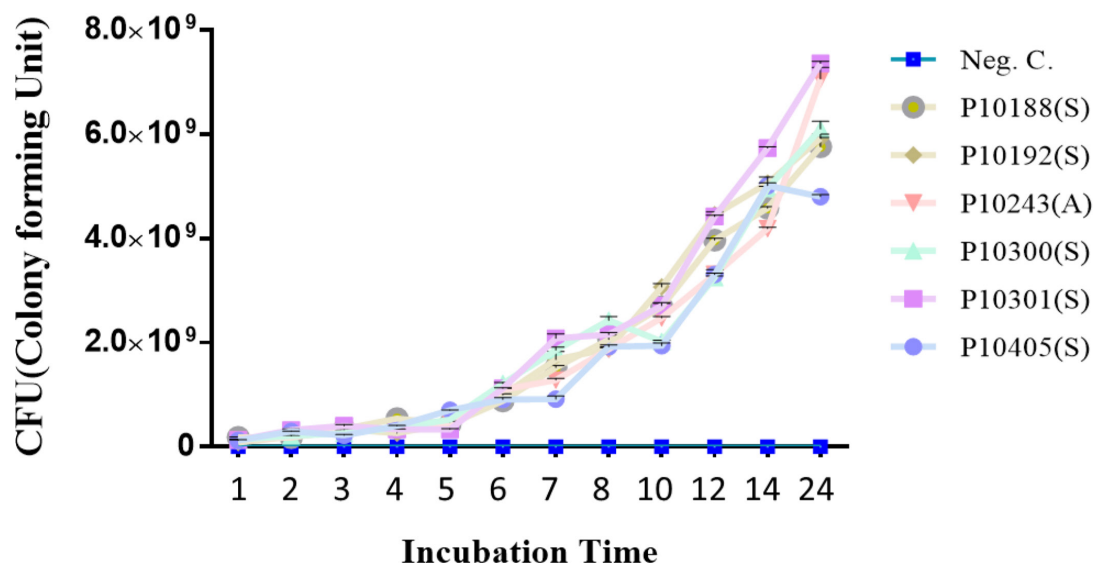


Fig. 2. Growth kinetics of the study isolates as CFU. ml⁻¹. Graph illustrates the growth of all isolates at different time intervals in hours. Similar growth pattern irrespective to symptomatic and asymptomatic isolates was observed. Error bars were calculated with SEM. The significance of data was analysed by using two-way ANOVA followed by Dunnett's multiple comparisons test by using GraphPad Prism version 6.0.0 for Windows, (GraphPad Software, San Diego, CA, USA, www.graphpad.com). (S) Symptomatic, (A) asymptomatic.

Growth kinetics and biofilm formation – Phenotypic and genetic observations

The asymptomatic isolate did not show any significant difference in growth kinetics or biofilm forming capacity in *in vitro* culture conditions as compared to symptomatic isolates (Fig. 2). All isolates have different biofilm formation ability irrespective of symptomatic or asymptomatic status. One symptomatic isolate (P10405) was observed to have maximum biofilm-formation property, followed by three other symptomatic isolates (P10192, P10300 and P10188), though it was not corroborated by severity of clinical symptoms in the host. The biofilm-formation ability of asymptomatic isolates was found similar to most of the symptomatic isolates (Fig. 3). We observed a comparable distribution of growth-related genes among the *E. coli* isolates in both symptomatic and asymptomatic bacteriuria conditions. This implies that isolates from asymptomatic bacteriuria cases have a potency of proper growth *in vitro* and *in vivo* but may not culminate into pathogenic potential in the host.

Sequence analysis for growth kinetics variation

It is known that variations in the TA system genes induce dormancy in cells. In our study, we checked the growth kinetics of all symptomatic and asymptomatic isolates. At optimal growing conditions, all the isolates showed equivalent growth characteristics, although there are sequence variations and its putative effect in TA system genes among the two groups (Fig. 2, Table 2).

Sequence analysis for biofilm formation

Biofilm formation is a well-proven survival strategy adapted by the bacteria during stress. In the current study, *in vitro* biofilm-formation capacity as well as the genomic variation in the biofilm-related genes were checked. As shown in the table (Table S2, Fig. 3), all genes related to pga operons, which is strongly related to biofilm formation were either mutated or absent in the asymptomatic isolate, but those were functional in all symptomatic isolates except in one (P10405). However, the presence of functional YidE (regulating biofilm formation) and other biofilm-formation genes (BssR, BssS, YjgK, cyclic-di-GMP-binding biofilm) in isolate P10405 resulted in the highest capacity of biofilm formation (Fig. 3). YidE was either non-functional or absent in the rest of the isolates (Table S2). Two chemotaxis proteins named methyl-accepting chemotaxis protein III (ribose and galactose chemoreceptor protein) and methyl-accepting chemotaxis protein IV (dipeptide chemoreceptor protein) were found non-functional in all symptomatic and asymptomatic isolates except P10405, which also support the maximum biofilm formation in *in vitro* experiments. PdeK (biofilm dispersal mediator) and YjgK (Biofilm regulation) were absent in P10301 (symptomatic isolate), but still it was observed to form biofilm *in vitro*, indicating non-essentiality of these genes in biofilm formation. Genes related to curli formation (critical for biofilm) were functional in all isolates except mutation of CsgE in a symptomatic isolate (P10188). Although CsgE and CsgF play important roles in the secretion of curli proteins, no effect was observed in the biofilm-forming phenotype in the isolates due to CsgE mutation [20], suggesting the existence of a compensatory pathway. Isolates P10188, P10192 and P10301 contained mutations of FliD (Fig. 7, Table 2 and Table S2), which did not show any variation in the biofilm-forming capacity at 48 hr in line with existing evidence [21].

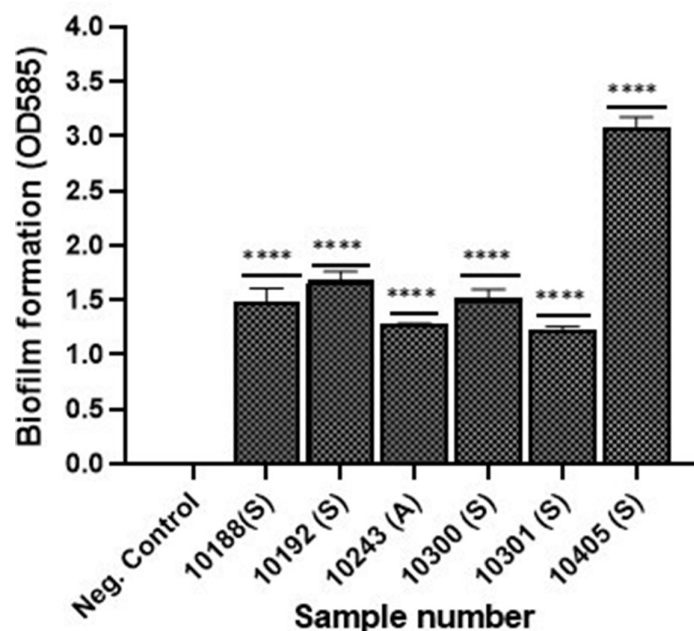


Fig. 3. Biofilm-formation ability of asymptomatic vs symptomatic isolates. Graph illustrated the biofilm-formation ability of all study isolates. P10405 shows the maximum biofilm formation with OD above 3.0 while ranging between 1.5–2.0 for the other five isolates. Error bars were calculated with SEM. The significance of data was analysed by using one-way ANOVA followed by Bartlett's multiple comparisons test by using GraphPad Prism version 6.0.0 for Windows, (GraphPad Software, San Diego, CA, USA, www.graphpad.com). (S) Symptomatic, (A) asymptomatic.

Genetic variation analysis of virulence genes

Three heat maps were constructed based on the virulence gene, antimicrobial resistance (AMR) and stress-management gene distribution among the symptomatic Vs asymptomatic isolates, comparing with reference genomes (*E. coli*_536, *E. coli*_CFT073, *E. coli*_K12 and *E. coli*_UTI89) relevant for UTI (Fig. 4). This comparative reference-based analysis suggested that no insertion/deletion mutations existed in the two categories of *E. coli* strains. But, a significant number of virulence and AMR genes reported in earlier evidence were not present in the reference *E. coli* genomes (*E. coli*_536, *E. coli*_CFT073, *E. coli*_K12 and *E. coli*_UTI89). So, for a deeper analysis, *de novo* genome assembly was subjected to RAST analysis with the reference *E. coli* taxonomy ID-562 [22–24] for identification of all relevant genetic loci. The genes related to various categories including virulence factor (other than previous), TA system, cell adhesion, chemotaxis, biofilm formation, curli, pili, and flagellar formation, iron accumulation and universal stress proteins were analysed. Apart from this, genes related to growth kinetics, biofilm formation and AST profile of all isolates were also checked to correlate the genomics with the phenotypic findings.

TA-system genes

Bacterial TA systems have largely contributed to the regulation of bacterial survival [25, 26]. Several types of TA systems are reported to be widely distributed in bacteria. In *E. coli*, approximately 36 types of TA systems have been reviewed till 2011 [26]. In this study, we had compared the genetic variations in the TA system among the isolates of *E. coli* from symptomatic and asymptomatic groups. Eight different types of fully functional TA systems named HipAB, YefM-YoeB, YeeU-YeeV (CbtA), YhaV-PrlF, ChpBS, HigAB, YgiUT and HicAB had been observed in the symptomatic isolates. These genes were present as toxin–antitoxin pairs. The isolates from symptomatic patients had more than one functional TA-system gene, while the isolate from asymptomatic patients did not have any fully functional TA system genes (Table 2). Moreover, the isolate from the later group had partially functional TA system with mutations in the toxin or antitoxin (Fig. 5). The toxins of HigAB and antitoxins of HipAB, yeeUV and YefM-YeoB of TA system were mutated in the asymptomatic isolate. Moreover, YgiUT system, HigAB system, HipAB, ChpBS system were only functional in some of the symptomatic isolates. HipAB has been reported to be a part of the type II TA system. The toxin (HipA) of HipAB inhibit cell growth by phosphorylating the aminoacyl-tRNA-Glu (tRNA^{Glu}) synthetase GltX. This leads to the accumulation of uncharged tRNA Glu; thereby triggering the stringent response [27, 28]. It has also been reported that the expression of HipA is lethal for the defective or HipB lacking bacterial strain, as the over expression of HipA gene is not tolerated by *E. coli* cell [29]. Therefore, it may be hypothesized that, in the absence of HipB (antitoxin), the cells with HipA might have entered into persistence mode and consequently become clinically asymptomatic. yefM/yoeB TA system plays an important role in the colonization of *E. coli* in the murine model and degradation of YefM increases persistence induction [30–32]. In the

Table 2. Genes with critical variation in asymptomatic (P10243) bacterial isolate. Functional changes are putative, logically derived from the nature of mutation and existing evidence of its effects in the published literature

S. no	Gene name	Variation	Function	Putative functional effect
1	Peptidyl-tRNA hydrolase (EC 3.1.1.29)	Deletion (107-194)	Sporulation	Functional change anticipated – deletion of 87 amino acids at C terminal only
3	Ferrichrome-iron receptor	Deletion (1-281; 444-760)	Iron accumulation	Functional change anticipated – gene neither has initiation nor C terminal
4	EfeB	AA substitution (A106V; K377E)	Iron accumulation	Functional change anticipated – amino acid substitution results in impaired folding of the protein [53, 54]
5	FeoB	AA substitution (S535N; I751V)	Iron accumulation	No functional change anticipated – amino acid substitution of similar kind
6	FetA	AA substitution (E2D)	Iron accumulation	No functional change anticipated – amino acid substitution of similar kind
7	PdeK	AA substitution (P446S)	c-di-GMP phosphodiesterase (EC 3.1.4.52)	Functional change anticipated – amino acid substitution results in impaired function of the protein [73]
8	HtrA	AA substitution (Q98L)	protease/chaperone protein	No functional change anticipated – amino acid substitution of similar kind
9	YjgK	AA substitution (A29V)	linked to biofilm formation	Functional change anticipated – amino acid substitution results in impaired folding of the protein [53]
10	CsgC	AA substitution (Q52K)	Putative curli production protein	Functional change anticipated – amino acid substitution results in impaired function of the protein [74]
11	FliK	AA substitution (V38L; T59A; S94L)	Flagellar protein	Functional change anticipated – amino acid substitution results in impaired function of the protein [75, 76]
12	FlgA	Deletion (1-63)	Flagellar protein	Functional change anticipated – loss of initiation site
13	FlgD	AA substitution (A29S)	Flagellar protein	No functional change anticipated – amino acid substitution of similar kind
14	CFA/I fimbrial subunit C usher protein	Deletion (618-625)	Type 1 fimbriae protein	Functional change anticipated – deletion of 7 amino acids at the C-terminal
15	Type-1 fimbrial protein, A chain	Absent	Type 1 fimbriae protein	Loss of function
16	FimE	Absent	Type 1 fimbriae protein	Loss of function
17	FimB	Absent	Type 1 fimbriae protein	Loss of function
18	FimG	Deletion (1-47; 137-167)	Type 1 fimbriae protein	Functional change anticipated – loss of initiation site
19	FimF	Absent	Type 1 fimbriae protein	Loss of function
20	FimC	Deletion (1-65; 145-241)	Type 1 fimbriae protein	Functional change anticipated – gene neither have initiation nor C terminal
21	FimH	Absent	Type 1 fimbriae protein	Toxin is mutated antitoxin is mutated antitoxin is mutated
22	HigAB	Mutation in Mutation in HipB	TA system	
23	HipAB	Mutation in YeeU	TA system	
24	YeeU/V	Mutation in YefM	TA system	
25	YefM-YeoB			

current study, mutation of YefM was observed in the asymptomatic isolate (P10243), which might have affected the colonization ability of cells resulting in a clinically asymptomatic isolate.

In earlier studies, toxin YeeV has been shown to affect cellular morphology and division, by inhibiting polymerization of two cytoskeleton proteins, MreB and FtsZ [33]. Antitoxin YeeU reduced the toxic effect of YeeV by binding with MerB and FtsZ (as



Fig. 4. Heatmap–insertion–deletion in xenobiotic resistance, virulence and stress-management genes of study isolates. Insertion and deletion in the genes that play a role in the bacterial xenobiotic resistance and virulence. (a) Xenobiotic resistance genes are broadly divided into two sub-categories – antibiotic resistance and metal resistance. Further, antibiotic resistance genes divided into EP and T – efflux pump and transport, OMP – outer membrane proteins and other genes those are reported to play a role in the antibiotic resistance but fully characterized, MR – metal resistance. (b) Virulence genes are further divided into sub-categories on the basis of their role in different cellular process. Ch include the genes that play a role in chemotaxis, BF includes biofilm-related genes, Co – colonization related genes, F represent filamentous-related genes and other genes that do not fall in the above categories but still have role in the virulence. (c) includes genes that are involved in the SM – stress management, SS – secretory system and TA – toxin antitoxin system. Insertion, deletion and both insertion and deletion, (S) symptomatic, (A)- asymptomatic.

competitive inhibitor) [34]. In the current study, antitoxin YeeU gene was absent in the asymptomatic isolate. This might have led to uninterrupted inhibition of cell division due to the toxin YeeV gene. VatA +VatP, and SatA +SatP critical for virulence, were reported in the UPEC isolates of other geographical regions, but were not observed in our study [35, 36].

In the present study, toxins of the HigAB TA system were also mutated in the asymptomatic isolate. It has been reported that HigAB TA system affects the virulence in *P. aeruginosa*, by regulating the production of pyocyanin (a toxin produced and secreted by *P. aeruginosa*), the siderophore pyochelin, surface motility (swarming) and biofilm formation [37, 38]. However, its presence among the UPEC isolate can be further looked for to establish any role in pathogenesis.

Sequence variation analysis for iron accumulation

Iron is an essential nutrient for the vast majority of bacterial species, playing many roles in the cellular mechanism, oxidative stress sensors (e.g. superoxide dismutases), haem or iron–sulphur cluster cofactors for different energy-generating and regulatory proteins etc. [39–42]. Iron deprivation can lead to death of an organism; while an excess of iron can cause damage to cellular machinery through free radical formation [43]. Variation in the genes related to ferrous transport (EfeB, FeoB, FeoC, FepE, FetA, in Feo, EfeUOB system), ferric ion transports (includes Fhu, Fep, Fec transporter system) and ferrichrome-iron receptor were observed in asymptomatic isolate as compared to symptomatic isolates (Fig. 6, Table 2 and Table S1). A key factor for regulating the intracellular iron availability is the action of various iron uptake systems ferrichrome-iron receptor (FIR). In our study, FIR mutations were observed in the asymptomatic isolate. This might have created iron deficiency-like conditions giving rise to loss of pathogenicity. Ferric enterobactin uptake protein (FepE), probable iron export ATP-binding protein (FetA), FeoB and FeoC and EfoB are reported to be involved in iron transport [44, 45]. EfeUOB system was functional among all symptomatic isolates, whereas the Feo system was functional in three symptomatic isolates (P10300, P10301 and P10405). Some synonymous amino acid substitutions were also observed in the asymptomatic isolate. All genes related to the Fhu system were functional other than FhuA. FhuA were mutated in all symptomatic as well as asymptomatic isolates. Another ferrichrome-iron receptor with similar function was present in all symptomatic isolates but mutated in the asymptomatic isolate. EfeB involves exogenous iron from

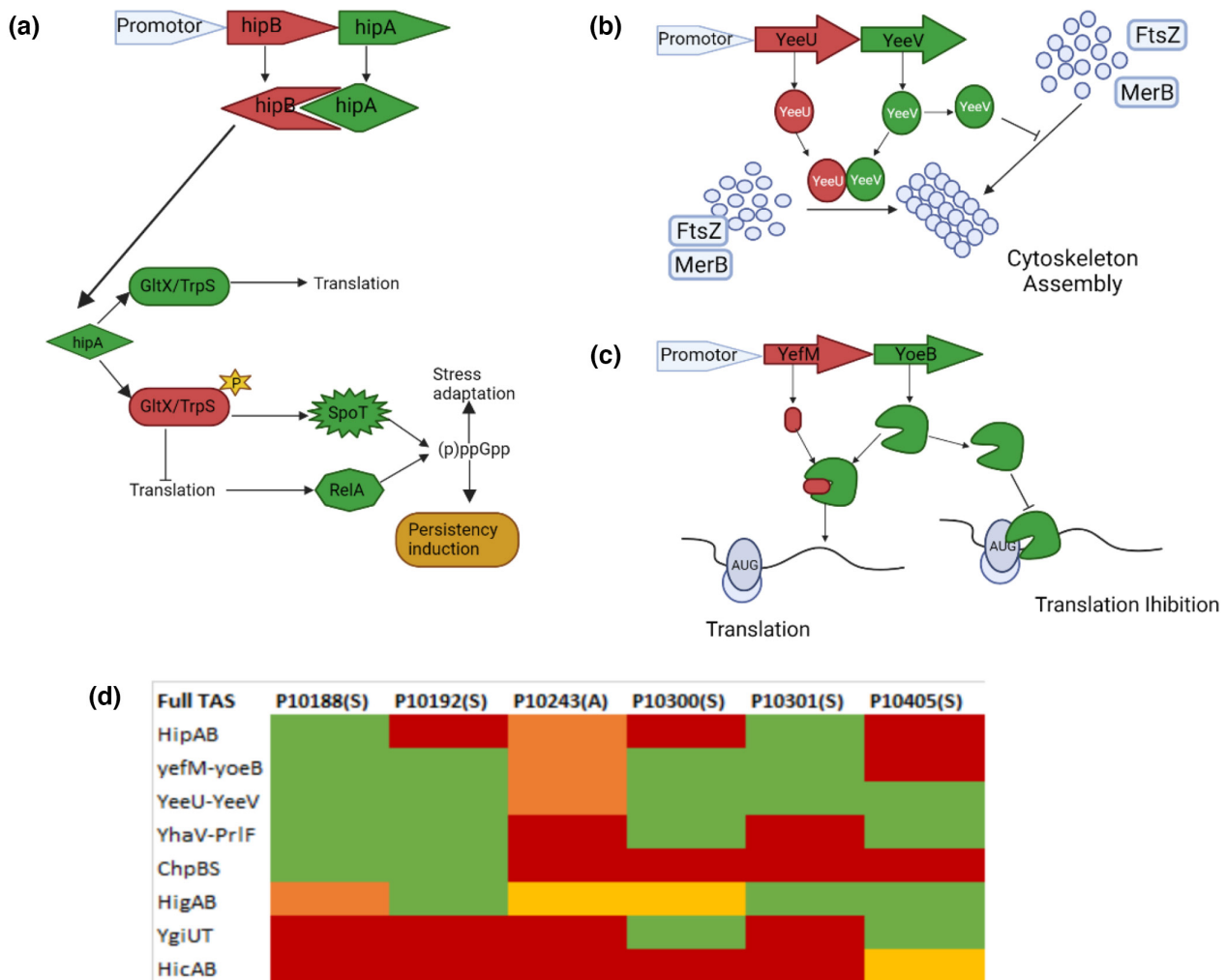


Fig. 5. Genetic variation in toxin-antitoxin genes of asymptomatic and symptomatic isolates. Figure illustrates a heat map of observed TA genes and work flow of HipAB, YeeU-YeeV and yefM-YoeB TA system. (a) Diagrammatic representation of HipAB and work flow of HipAB TA system, (b) work flow of YeeU-YeeV TA system, (c) work flow of yefM-YoeB TA system. The asymptomatic isolate has mutated antitoxins, which are denoted by red while functional toxin is denoted by grey. (d) Heat map of present TA genes in each isolate, heat map colour code is (S) symptomatic, (A) asymptomatic. Both TA are functional, toxin functional while antitoxin non-functional, antitoxin functional while toxin non-functional, both TA are non-functional. Figure was edited and remodelled based on study data from KEGG (<https://www.genome.jp/kegg/kegg1.html>) by using BioRender.com.

haem, and displays peroxidase activity *in vitro*, FeoB and FeoC involved in the ferrous iron uptake and regulation [46–49]. During infection the iron remains bound with haem or circulating proteins (transferrin or lactoferrin) and is not available freely. Hence, bacteria may face difficulty in iron uptake inside the host body [50, 51]. Pathogens adopt various strategies during infection to uptake iron in different forms. Siderophores are commonly used by the bacteria for iron uptake followed by uptake of Fe²⁺ by Feo system and iron by haem [52, 53]. In the current study, absence or mutated allele of EfeB (iron uptake from haem), FeoB and FeoC (ferrous iron by Feo system) and ferrichrome-iron receptor (siderophores mediated iron accumulation) putatively reduced the iron accumulation capacity in the asymptomatic isolate (Fig. 6, Table 2 and Table S2). The nature of mutations in EfeB as single mutations changing the amino acids, may have resulted in impaired folding or altered kinetic properties ([53, 54]. Fep transport system was only functional in two symptomatic isolates. As per the published literatures, iucD has an important role in the virulence of *E. coli* [55]. However, it was found absent in all the isolates in our study.

Collectively, it can be concluded that mutation in iron accumulation genes in asymptomatic isolate of *E. coli* may lead to lack of virulence *in vivo*. Apart from these genes, few mutations were observed in the symptomatic isolates affecting minor pathways. Overall it indicates that existence of any one functional iron-accumulation pathway is enough for survival and virulence in the invading pathogen.



Fig. 6. Heat map of genetic variation in iron accumulation genes of studied isolates. Genes involved in iron accumulation are divided into two categories, ferrous and ferric iron transport. Asymptomatic isolate is showing sequence variation in both pathways giving rise to putative functional change. Represent fully functional genes, denotes deletion, denotes amino acid substitution, FIR – errichrome-iron receptor.

Apart from the above mentioned genes, *hlyA* and *cnf-1* have been shown as a virulence marker for causation of UTI in the published literature [56] and these were found absent in the studied isolates.

Sequence variation of curli, flagella, chemotaxis, pilus and Fimbriae

It has been known that, outer membrane protein and various structural factors such as curli, fimbriae, pilli, flagella etc. play an important role in cell adhesion, colonization, invasion, evasion of host immune system and intracellular survival. Curli are the proteinaceous extracellular fibres of the bacteria involved in cell-surface adhesion and cell-to-cell attachment, community behaviour and host colonization [57–60]. In the current study, genes (*CsgA/B/C/D/E/F/G*) as well as transcriptional activators were observed to be functional in all isolates irrespective of their clinical presentation and biofilm formation ability.

Fifty-two different types of flagellar genes that includes *FliD/E/F/G/H/I/J/K/L/M/N/O/P/Q/R/S/T/Z*, *FlgA/B/C/D/E/F/G/H/I/J/K/L/M/N*, *MotA*, *MotB*, *FlhA/B/C/D/E*, and flagellar RNA polymerase sigma factor, RNA polymerase sigma-54 factor *RpoN*, *Flk*, *FimA/B/C/D/E/F/G/H/X* were studied in all the isolates. Fifteen (*CFA/I*, *TFP-A*, *FimE*, *FimB*, *FimH*, *FimC*, *FimG*, *FimF*, *FlgA*, *MotB*, *FlgE*, *FlhE*, *FimF*, *FimD* and *MACP-3*) out of 52 genes were non-functional in the asymptomatic isolate. *FlgE* gene was mutated in all the isolates, whereas *Flil*, RNA polymerase sigma factor, *fliD* and *flhE* were observed to be mutated in many isolates (Fig. 7, Table S2). Fimbriae antigen *FimX* is absent in all isolates. Flagellum is primarily a motility organelle that enables movement and chemotaxis, participates in biofilm formation, protein export and adhesion [61]. Flagellar motility benefits the bacteria to move toward favourable environmental conditions, with regulation of gene expression in response to external environments [62, 63]. Flagellar-mediated motility is considered as a virulence factor for several gram-negative pathogens along with *E. coli* [44, 45, 64–66]. In the present study, functional variation in flagellar genes were found in the asymptomatic isolate (Fig. 7, Table S2). Since flagellar-mediated motility is required for host–pathogen interaction, these variations might have rendered non-pathogenicity.

Chemotaxis also plays an important role in the biofilm formation. Hence, genes related to chemotaxis were also checked for variations. Methyl-accepting chemotaxis protein III (*MCP-III*) (serine chemoreceptor protein) and methyl-accepting chemotaxis protein IV (serine chemoreceptor protein) were either absent or mutated in all isolates except one symptomatic isolate (P10405), rendering high biofilm-forming capacity as observed experimentally. Fimbriae and pilus play an important role in bacterial adherence, colonization and pathogenicity. Genes related to fimbriae and pilus were also checked for any variations. No variation was observed in the genes including *CFA/I* fimbrial auxiliary subunit, *CFA/I* fimbrial chaperone, *CFA/I* fimbrial major subunit and *CFA/I* fimbrial minor adhesin. *CFA/I* fimbrial subunit C usher protein, type-1 fimbrial protein A chain (*TFP-A*), *FimE*, *FimB*, *FimG* and *FimF* were mutated in asymptomatic isolate (Fig. 7, Table 2 and Table S2). All symptomatic isolates had functional alleles of these fimbriae genes except *FimB* in P10188 and P10192 (N-terminal domain deletion). This indicates a non-essential role of the N-terminal domain of *FimB* in fimbriae development. Type-1 fimbriae in *E. coli* plays a crucial role in the adhesion to bladder and thereby establishing the UTI infection [67–70]. In the present study, the asymptomatic isolate did not have a functional allele of type-1 fimbrial protein. This might be one of the probable causes of lack of symptoms *in vivo*, in spite of normal growth kinetics of the isolate in *in vitro* conditions. All isolates carry functional allele of type-IV pilin (*PilA/B/C/M/N/O/P/Q/T*) along with other regulatory genes. The role of *pap* genes is well proven in the adhesion and pathogenicity of *E. coli* in case of

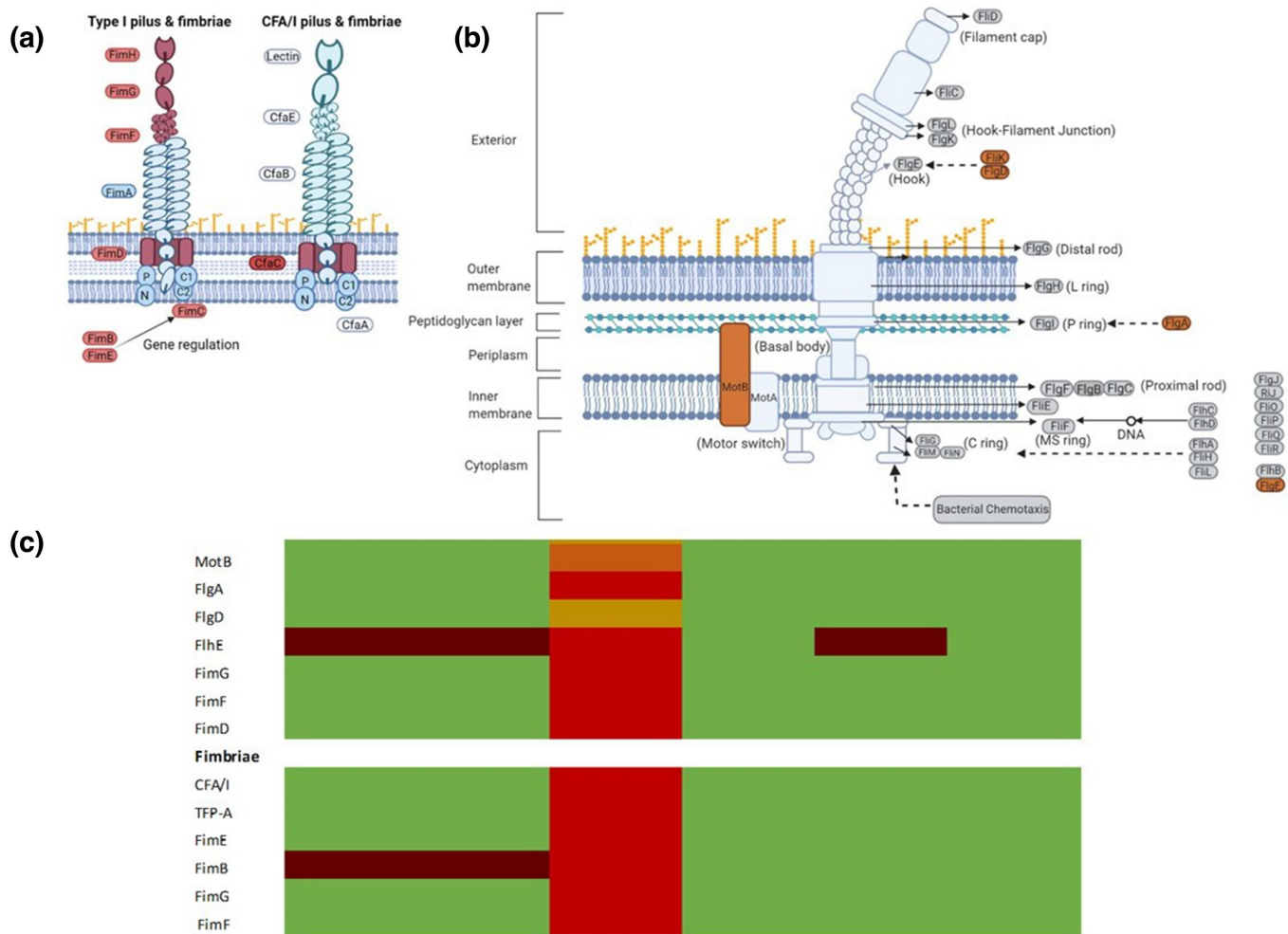


Fig. 7. Flagellar, chemotaxis and fimbriae formation. Genes involved in the formation of flagella, chemotaxis and fimbriae formation are showing variation in the asymptomatic isolate as compared to symptomatic isolates. The flagellar genes are observed to have maximum genetic variation in the asymptomatic isolate as compared to the symptomatic isolate. Genes related to fimbriae formation are mostly mutated in asymptomatic isolates. (a) Flagella formation in asymptomatic isolate, (b) pilus and fimbriae formation in asymptomatic isolate. Red represents mutation and grey denotes functional allele. (c) Heat map of genetic variation in asymptomatic vs symptomatic isolate. (S) Symptomatic, (A) asymptomatic, functional gene, putative non-effective mutation (putative functional allele), denotes amino acid substitution, gene have putative effective mutation (putative non-functional allele). CFA/I- CFA/I fimbrial subunit C usher protein and TFP-A-type-1 fimbrial protein, A chain. Figure was edited and remodelled based on study data, from KEGG (<https://www.genome.jp/kegg/kegg1.html>) by using BioRender.com.

UTI [59]. In the current study, pap genes involved in the formation of P pili were absent. UPEC from the hospital setting are known to possess Fim-like tyrosine recombinases that includes FimA, FimH, FimC, FimB, FimX, IpuA, IpuB, etc. [17, 18, 71]. In the current study, FimB, and FimX, were present in the isolates from both symptomatic and asymptomatic, whereas FimH and FimC were present only in the symptomatic isolates. This emphasizes a possible role of FimB and FimX genes in survival and FimH and FimC in the causation of disease (Fig. 8). It has been reported that UPEC from hospital settings contain various other virulence factors like PapA, PapC, PapE PapG, HyxA, hyxB, hyxR, ibeA, trat, fyuA, hlyA, cnf1, iha, iucD, draA, sfaD, neuB, KfiC, iutA, iroN and isrC [17, 35, 72]. But in our study, all genes were absent. This difference might have arisen from different types of antibiotics selection pressure in hospital settings as compared to community settings.

CONCLUSION

This study was carried out to determine the differentiating genomic characteristics of symptomatic and asymptomatic bacteriuria with respect to virulence genes, biofilm-formation genes and iron transport/regulation genes. The asymptomatic isolate had significant genetic variation in multiple virulence gene loci as compared to symptomatic isolates. These variations were observed mainly in the TA systems genes (HipAB, yefM-yoeB and YeeU-YeeV), iron accumulation (EfeB, FeoB, FeoC, FepE and FetA), flagellar development

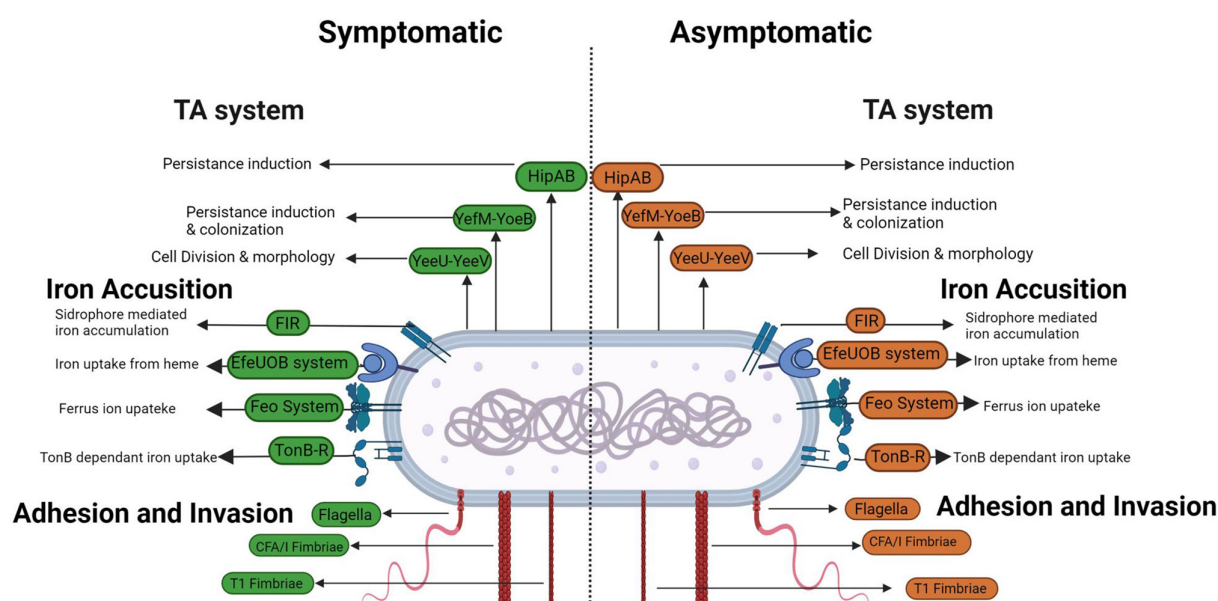


Fig. 8. Comparison of asymptomatic vs asymptomatic groups. Summary figure illustrating the difference between symptomatic and asymptomatic isolates. Green represents functional gene of particular category, and red denotes mutation in impaired function or loss of function of one or more genes in a particular category.

genes (FliK, MotB, FlgA, FlgD, FlgE, FlhE, FimG, FimF, FimD), chemotaxis (MACP-3) and fimbriae formation (CFA/I, TFP-A, FimE, FimB, FimG and FimF) genes (Fig. 8). This indicates a possibility of emergence of these mutations either by gene transfer or deletion/insertion that may have occurred due to environmental stress or drug pressure in the community. Significant sequence variations putatively resulting in enhanced pathogenicity in symptomatic isolates or suppressed virulence in the asymptomatic isolate were observed. Further analysis in our study cohort on similar mutations rendering specific virulence phenotype is anticipated to strengthen our hypothesis. The intriguing observations in the present study generates ample opportunity for intensive functional genetic studies to further illustrate the essentiality of these genes for a symptomatic phenotype in bacteriuria cases.

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Conflicts of interest

The authors declare that there are no conflicts of interest.

References

1. Ndamason LM, Marbou WJ, Kuete V. Urinary tract infections, bacterial resistance and immunological status: a cross sectional study in pregnant and non-pregnant women at Mbouda Ad-Lucem Hospital. *Afr Health Sci* 2019;19:1525–1535.
2. Le J, Briggs GG, McKeown A, Bustillo G. Urinary tract infections during pregnancy. *Ann Pharmacother* 2004;38:1692–1701.
3. Akerlele J, Abhulimen P, Okonofua F. Prevalence of asymptomatic genital infection among pregnant women in Benin City, Nigeria. *Afr J Reprod Health* 2002;6:93–97.
4. Onuh SUO, Igberase G, Azikem M, Okpere E. Microbiological isolates and sensitivity pattern of Urinary Tract Infection in pregnancy in Benin City, Nigeria. *Ebony Medical Journal* 2009;5:48–52.
5. Angelescu K, Nussbaumer-Streit B, Sieben W, Scheibler F, Gartlehner G. Benefits and harms of screening for and treatment of asymptomatic bacteriuria in pregnancy: a systematic review. *BMC Pregnancy Childbirth* 2016;16:336.
6. Hooton TM, Stamm WE. Diagnosis and treatment of uncomplicated urinary tract infection. *Infect Dis Clin North Am* 1997;11:551–581.
7. Leimbach A, Hacker J, Dobrindt U. *E. coli* as an all-rounder: the thin line between commensalism and pathogenicity. *Curr Top Microbiol Immunol* 2013;358:3–32.
8. Wood JL, Osman A, Wade SA. An efficient, cost-effective method for determining the growth rate of sulfate-reducing bacteria using spectrophotometry. *MethodsX* 2019;6:2248–2257.
9. Krueger F. Trim Galore!: A wrapper tool around Cutadapt and FastQC to consistently apply quality and adapter trimming to FastQ files. *Babraham Bioinformatics*; 2021. https://www.bioinformatics.babraham.ac.uk/projects/trim_galore/
10. Langmead B, Salzberg SL. Fast gapped-read alignment with bowtie 2. *Nat Methods* 2012;9:357–359.

11. Li H, Handsaker B, Wysoker A, Fennell T, Ruan J, et al. The Sequence Alignment/Map format and SAMtools. *Bioinformatics* 2009;25:2078–2079.
12. Li H. A statistical framework for SNP calling, mutation discovery, association mapping and population genetical parameter estimation from sequencing data. *Bioinformatics* 2011;27:2987–2993.
13. Cingolani P, Platts A, Wang LL, Coon M, Nguyen T, et al. A program for annotating and predicting the effects of single nucleotide polymorphisms, SnpEff: SNPs in the genome of *Drosophila melanogaster* strain w1118; iso-2; iso-3. *Fly (Austin)* 2012;6:80–92.
14. Kryger J, Burleigh A, Christensen M, Hopkins W. Genetic evaluation of *E. coli* strains isolated from asymptomatic children with neurogenic bladders. *Int J Chronic Dis* 2015;2015:206570.
15. Mabbett AN, Ulett GC, Watts RE, Tree JJ, Totsika M, et al. Virulence properties of asymptomatic bacteriuria *Escherichia coli*. *Int J Med Microbiol* 2009;299:53–63.
16. Srivastava S, Agarwal J, Mishra B, Srivastava R. Virulence versus fitness determinants in *Escherichia coli* isolated from asymptomatic bacteriuria in healthy nonpregnant women. *Indian J Med Microbiol* 2016;34:46–51.
17. Bateman SL, Stapleton AE, Stamm WE, Hooton TM, Seed PC. The type 1 pili regulator gene fimX and pathogenicity island PAI-X as molecular markers of uropathogenic *Escherichia coli*. *Microbiology (Reading)* 2013;159:1606–1617.
18. Hannan TJ, Mysorekar IU, Chen SL, Walker JN, Jones JM, et al. LeuX tRNA-dependent and -independent mechanisms of *Escherichia coli* pathogenesis in acute cystitis. *Mol Microbiol* 2008;67:116–128.
19. Najafi A, Hasanpour M, Askary A, Aziemzadeh M, Hashemi N. Distribution of pathogenicity island markers and virulence factors in new phylogenetic groups of uropathogenic *Escherichia coli* isolates. *Folia Microbiol (Praha)* 2018;63:335–343.
20. Evans ML, Chapman MR. Curli biogenesis: order out of disorder. *Biochim Biophys Acta* 2014;1843:1551–1558.
21. Ratthawongjirakul P, Thongkerd V, Chaicumpa W. The impacts of a fliD mutation on the biofilm formation of *Helicobacter pylori*. *Asian Pac J Trop Biomed* 2016;6:1008–1014.
22. Aziz RK, Bartels D, Best AA, DeJongh M, Disz T, et al. The RAST Server: rapid annotations using subsystems technology. *BMC Genomics* 2008;9:75.
23. Brettin T, Davis JJ, Disz T, Edwards RA, Gerdes S, et al. RASTtk: a modular and extensible implementation of the RAST algorithm for building custom annotation pipelines and annotating batches of genomes. *Sci Rep* 2015;5:8365.
24. Overbeek R, Olson R, Pusch GD, Olsen GJ, Davis JJ, et al. The SEED and the rapid annotation of microbial genomes using Subsystems Technology (RAST). *Nucleic Acids Res* 2014;42:D206–14.
25. Engelberg-Kulka H, Amitai S, Kolodkin-Gal I, Hazan R. Bacterial programmed cell death and multicellular behavior in bacteria. *PLoS Genet* 2006;2:e135.
26. Yamaguchi Y, Park JH, Inouye M. Toxin-antitoxin systems in bacteria and archaea. *Annu Rev Genet* 2011;45:61–79.
27. Germain E, Castro-Roa D, Zenkin N, Gerdes K. Molecular mechanism of bacterial persistence by HipA. *Mol Cell* 2013;52:248–254.
28. Kaspy I, Rotem E, Weiss N, Ronin I, Balaban NQ, et al. HipA-mediated antibiotic persistence via phosphorylation of the glutamyl-tRNA-synthetase. *Nat Commun* 2013;4:3001.
29. Moyed HS, Broderick SH. Molecular cloning and expression of hipA, a gene of *Escherichia coli* K-12 that affects frequency of persistence after inhibition of murein synthesis. *J Bacteriol* 1986;166:399–403.
30. Grady R, Hayes F. Axe-Txe, a broad-spectrum proteic toxin-antitoxin system specified by a multidrug-resistant, clinical isolate of *Enterococcus faecium*. *Mol Microbiol* 2003;47:1419–1432.
31. Nieto C, Cherny I, Khoo SK, de Lacoba MG, Chan WT, et al. The yefM-yoeB toxin-antitoxin systems of *Escherichia coli* and *Streptococcus pneumoniae*: functional and structural correlation. *J Bacteriol* 2007;189:1266–1278.
32. Ramisetty BCM, Ghosh D, Roy Chowdhury M, Santhosh RS. What is the link between stringent response, endoribonuclease encoding type II toxin-antitoxin systems and persistence? *Front Microbiol* 2016;7:1882.
33. Tan Q, Awano N, Inouye M. YeeV is an *Escherichia coli* toxin that inhibits cell division by targeting the cytoskeleton proteins, FtsZ and MreB. *Mol Microbiol* 2011;79:109–118.
34. Masuda H, Tan Q, Awano N, Wu KP, Inouye M. YeeU enhances the bundling of cytoskeletal polymers of MreB and FtsZ, antagonizing the CbtA (YeeV) toxicity in *Escherichia coli*. *Mol Microbiol* 2012;84:979–989.
35. Ballesteros-Monreal MG, Arenas-Hernández MM, Enciso-Martínez Y, Martínez-de la Peña CF, Rocha-Gracia RDC, et al. Virulence and resistance determinants of uropathogenic *Escherichia coli* strains isolated from pregnant and non-pregnant women from two states in Mexico. *Infect Drug Resist* 2020;13:295–310.
36. Xicohtencatl-Cortes J, Cruz-Córdova A, Cázares-Domínguez V, Escalona-Venegas G, Zavala-Vega S, et al. Uropathogenic *Escherichia coli* strains harboring *tosA* gene were associated to high virulence genes and a multidrug-resistant profile. *Microb Pathog* 2019;134:103593.
37. Fernández-García L, Blasco L, Lopez M, Bou G, García-Contreras R, et al. Toxin-antitoxin systems in clinical pathogens. *Toxins (Basel)* 2016;8:E227.
38. Wood TL, Wood TK. The HigB/HigA toxin/antitoxin system of *Pseudomonas aeruginosa* influences the virulence factors pyochelin, pyocyanin, and biofilm formation. *Microbiologyopen* 2016;5:499–511.
39. Braun V, Hantke K. Recent insights into iron import by bacteria. *Curr Opin Chem Biol* 2011;15:328–334.
40. Chu BC, Garcia-Herrero A, Johanson TH, Krewulak KD, Lau CK, et al. Siderophore uptake in bacteria and the battle for iron with the host; a bird's eye view. *Biomaterials* 2010;23:601–611.
41. Hassan HM, Schrum LW. Roles of manganese and iron in the regulation of the biosynthesis of manganese-superoxide dismutase in *Escherichia coli*. *FEMS Microbiol Rev* 1994;14:315–323.
42. C. RR. *Iron Metabolism: From Molecular Mechanisms to Clinical Consequences*. Chichester, West Sussex: UK John Wiley and Sons Ltd; 2009.
43. Touati D. Iron and oxidative stress in bacteria. *Arch Biochem Biophys* 2000;373:1–6.
44. Allen-Vercoe E, Woodward MJ. The role of flagella, but not fimbriae, in the adherence of *Salmonella enterica* serotype Enteritidis to chick gut explant. *J Med Microbiol* 1999;48:771–780.
45. Marchetti M, Sirard JC, Sansonetti P, Pringault E, Kernéis S. Interaction of pathogenic bacteria with rabbit appendix M cells: bacterial motility is a key feature in vivo. *Microbes Infect* 2004;6:521–528.
46. Kammler M, Schön C, Hantke K. Characterization of the ferrous iron uptake system of *Escherichia coli*. *J Bacteriol* 1993;175:6212–6219.
47. Kim H, Lee H, Shin D. The FeoC protein leads to high cellular levels of the Fe(II) transporter FeoB by preventing FtsH protease regulation of FeoB in *Salmonella enterica*. *J Bacteriol* 2013;195:3364–3370.
48. Liu X, Du Q, Wang Z, Zhu D, Huang Y, et al. Crystal structure and biochemical features of EfeB/YcdB from *Escherichia coli* O157: ASP235 plays divergent roles in different enzyme-catalyzed processes. *J Biol Chem* 2011;286:14922–14931.
49. Weaver EA, Wyckoff EE, Mey AR, Morrison R, Payne SM. FeoA and FeoC are essential components of the *Vibrio cholerae* ferrous iron uptake system, and FeoC interacts with FeoB. *J Bacteriol* 2013;195:4826–4835.
50. Cornelissen CN, Sparling PF. Iron piracy: acquisition of transferrin-bound iron by bacterial pathogens. *Mol Microbiol* 1994;14:843–850.
51. Finkelstein RA, Sciortino CV, McIntosh MA. Role of Iron in Microbe-Host Interactions. *Rev Infect Dis* 1983;5:S759–S777.
52. Andrews S, Norton I, Salunkhe AS, Goodluck H, Aly WS, et al. Control of iron metabolism in bacteria. *Met Ions Life Sci* 2013;12:203–239.

53. Imoto T, Yamada H, Yasukochi T, Yamada E, Ito Y, et al. Point mutation of alanine (31) to valine prohibits the folding of reduced lysozyme by sulfhydryl–disulfide interchange. *Protein Eng Des Sel* 1987;1:333–338.
54. Farrés J, Wang X, Takahashi K, Cunningham SJ, Wang TT, et al. Effects of changing glutamate 487 to lysine in rat and human liver mitochondrial aldehyde dehydrogenase. A model to study human (Oriental type) class 2 aldehyde dehydrogenase. *J Biol Chem* 1994;269:13854–13860.
55. Gao Q, Wang X, Xu H, Xu Y, Ling J, et al. Roles of iron acquisition systems in virulence of extraintestinal pathogenic *Escherichia coli*: salmochelin and aerobactin contribute more to virulence than heme in a chicken infection model. *BMC Microbiol* 2012;12:143.
56. López-Banda DA, Carrillo-Casas EM, Leyva-Leyva M, Orozco-Hoyuela G, Manjarrez-Hernández AH, et al. Identification of virulence factors genes in *Escherichia coli* isolates from women with urinary tract infection in Mexico. *Biomed Res Int* 2014;2014:959206.
57. Gophna U, Barlev M, Seijffers R, Oelschlager TA, Hacker J, et al. Curli fibers mediate internalization of *Escherichia coli* by eukaryotic cells. *Infect Immun* 2001;69:2659–2665.
58. Gophna U, Oelschlaeger TA, Hacker J, Ron EZ. Role of fibronectin in curli-mediated internalization. *FEMS Microbiol Lett* 2002;212:55–58.
59. Zogaj X, Bokranz W, Nimtz M, Römling U. Production of cellulose and curli fimbriae by members of the family *Enterobacteriaceae* isolated from the human gastrointestinal tract. *Infect Immun* 2003;71:4151–4158.
60. Zogaj X, Nimtz M, Rohde M, Bokranz W, Römling U. The multicellular morphotypes of *Salmonella typhimurium* and *Escherichia coli* produce cellulose as the second component of the extracellular matrix. *Mol Microbiol* 2001;39:1452–1463.
61. Haiko J, Westerlund-Wikström B. The role of the bacterial flagellum in adhesion and virulence. *Biology (Basel)* 2013;2:1242–1267.
62. Fenchel T. Microbial behavior in a heterogeneous world. *Science* 2002;296:1068–1071.
63. Soutourina OA, Bertin PN. Regulation cascade of flagellar expression in Gram-negative bacteria. *FEMS Microbiol Rev* 2003;27:505–523.
64. Black RE, Levine MM, Clements ML, Hughes TP, Blaser MJ. Experimental *Campylobacter jejuni* infection in humans. *J Infect Dis* 1988;157:472–479.
65. Krukonis ES, DiRita VJ. From motility to virulence: Sensing and responding to environmental signals in *Vibrio cholerae*. *Curr Opin Microbiol* 2003;6:186–190.
66. La Ragione RM, Sayers AR, Woodward MJ. The role of fimbriae and flagella in the colonization, invasion and persistence of *Escherichia coli* O78:K80 in the day-old-chick model. *Epidemiol Infect* 2000;124:351–363.
67. Hagberg L, Jodal U, Korhonen TK, Lidin-Janson G, Lindberg U, et al. Adhesion, hemagglutination, and virulence of *Escherichia coli* causing urinary tract infections. *Infect Immun* 1981;31:564–570.
68. Iwahi T, Abe Y, Nakao M, Imada A, Tsuchiya K. Role of type 1 fimbriae in the pathogenesis of ascending urinary tract infection induced by *Escherichia coli* in mice. *Infect Immun* 1983;39:1307–1315.
69. Källenius G, Möllby R, Winberg J. *In vitro* adhesion of uropathogenic *Escherichia coli* to human periurethral cells. *Infect Immun* 1980;28:972–980.
70. Kuch B, Pál T, Emödy L. New haemagglutinating fimbriae on *Escherichia coli* strains isolated from urine. *Acta Microbiol Acad Sci Hung* 1980;27:317–323.
71. Bryan A, Roesch P, Davis L, Moritz R, Pellett S, et al. Regulation of type 1 fimbriae by unlinked FimB- and FimE-like recombinases in uropathogenic *Escherichia coli* strain CFT073. *Infect Immun* 2006;74:1072–1083.
72. Grude N, Potaturkina-Nesterova NI, Jenkins A, Strand L, Nowrouzian FL, et al. A comparison of phylogenetic group, virulence factors and antibiotic resistance in Russian and Norwegian isolates of *Escherichia coli* from urinary tract infection. *Clin Microbiol Infect* 2007;13:208–211.
73. Ozata M, Suzuki S, Takeda T, Malkin DG, Miyamoto T, et al. Functional analysis of a proline to serine mutation in codon 453 of the thyroid hormone receptor beta 1 gene. *J Clin Endocrinol Metab* 1995;80:3239–3245.
74. Singh R, Rathore YS, Singh NS, Peddada N, et al. Substitution of glutamate residue by lysine in the dimerization domain affects DNA binding ability of HapR by inducing structural deformity in the DNA binding domain. *PLoS One* 2013;8:e76033.
75. Kazemi-Esfarjani P, Beitel LK, Trifiro M, Kaufman M, Rennie P, et al. Substitution of valine-865 by methionine or leucine in the human androgen receptor causes complete or partial androgen insensitivity, respectively with distinct androgen receptor phenotypes. *Mol Endocrinol* 1993;7:37–46.
76. Rosen HN, Murrell JR, Liepnieks JJ, Benson MD, Cody V, et al. Threonine for alanine substitution at position 109 of transthyretin differentially alters human transthyretin's affinity for iodothyronines. *Endocrinology* 1994;134:27–34.

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